



FIGURE 11.18 Beauty is in the eye of the beholder. Many cultures, including Europe in earlier centuries, have had a standard of female beauty that is noticeably plumper than that of Western society today. Presumably, a society with no pressure for extreme thinness is also a culture with little likelihood of anorexia nervosa.

postage stamps, for fear that the glue might contain some tiny fraction of a calorie. A mere lack of hunger could hardly explain such behavior.

A better description of anorexia nervosa is that it reflects a “pathological fear of fatness.” Even when anorexic women become painfully thin, they often describe themselves as “looking fat” and “needing to lose weight” (Figure 11.19).

What motivates someone to become anorexic? The motivations are complex and vary from one individual to another. First, many women who become anorexic have always prized self-control. Their massive weight loss demonstrates extreme self-control and thereby raises their self-esteem. Also, maintaining a dangerously low weight can be a way of rebelling quietly and attracting attention. Before the onset of the disorder, most anorexic girls are described as having been obedient, high-achieving perfectionists (Bruch, 1980; Goldstein, 1981). Their severe weight loss attracts attention and gives them prestige.

Bulimia

Other people, again mostly young women, starve themselves at times but occasionally throw themselves into an eating binge. They may consume as many as 20,000 calories at a time (Schlesier-Stropp, 1984)—the equivalent of about 30 Big Macs, 10 helpings of French fries, and 10 chocolate milkshakes. Some, but not all, force themselves to vomit or use laxatives after gorging on these enormous meals. People who *alternate between self-starvation and excessive eating* are said to suffer from **bulimia** (literally,

“ox hunger”). We might imagine that people who go on eating binges might starve themselves for a while to make up for it. According to Janet Polivy and Peter Herman (1985), however, the causation goes in the other direction: It is the dieting that causes the binges. Bulimic people starve themselves for a while, fight their persistent feelings of hunger for a while, and then go on an eating binge.

Most people with bulimia have low self-esteem, a tendency toward depression, and a history of growing up in a somewhat troubled family (Fairburn, Welch, Doll, Davies, & O'Connor, 1997). Of course, so do many other people who do not develop bulimia. Some people with bulimia are thin (as with anorexia), but others fluctuate around a normal or high body weight. One study of women who sought psychological help for bulimia found that, 6 years after the end of treatment, 60% no longer had an eating disorder, 30% still had a moderate problem, and 10% either had serious disorders or had died (Fichter & Quadflieg, 1997).

Eating disorders including anorexia and bulimia increased in prevalence in the United States during the 1980s and 1990s, especially among women. Meanwhile, most surveys also found an increasing percentage of young women who report feeling dissatisfied with their appearance, especially those who think they are too heavy (Feingold & Mazzella, 1998). The pressure on women to try to look thin is probably a major contributor to both anorexia and bulimia.



FIGURE 11.19 A fun house mirror causes a temporary distortion of anyone's body appearance. People with anorexia nervosa experience a similar distortion of body image at all times, seeing themselves as much fatter than they really are.

THE MESSAGE

The Complexities of Hunger

The research on anorexia and bulimia underscores the idea that our motivations are controlled by a complex mixture of physiological, social, and cognitive forces. People become overweight (or falsely perceive themselves as overweight) for a variety of reasons, relating to everything from genetics to culture, and they then try to lose weight mostly for social reasons, such as trying to look attractive. Sometimes, the physiological factors and the social factors collide, as when normal-weight people try to make themselves thinner and thinner.

The overall point is this: All our motivations interact and combine. How much we eat and what and when we eat depend not only on our need for food, but also on social needs and the need for self-esteem.

SUMMARY

★ **Food selection.** Food preferences can be altered by changes in body chemistry, such as a deficiency of salt. Other things being equal, we tend to prefer familiar foods. We avoid foods that have been associated with something nauseating or repulsive and prefer foods previously associated with other good-tasting foods. (page 400)

★ **Short-term regulation of hunger.** Meals are ended by several mechanisms, principally distension of the stomach and intestines. Hunger resumes when the cells begin to receive less glucose and other nutrients. The hormones insulin and glucagon regulate the flow of nutrients from the blood to storage and from there back into the blood. (page 402)

★ **Long-term regulation of hunger.** An individual meal can be larger or smaller than necessary to provide the energy that the body needs. The body's fat cells secrete the hormone leptin in proportion to their mass; an increase of leptin decreases hunger, and a decrease of leptin increases hunger. (page 405)

★ **Causes of being overweight.** People become overweight for many reasons, but emotional difficulties are rarely a major factor. Some people are predisposed to obesity for genetic reasons; whether they become obese or how obese they become depends on what kinds of food their culture provides. Among the mechanisms leading to obesity are low energy output and insensitivity to leptin. (page 406)

★ **Weight-loss techniques.** People in our society resort to a variety of strategies to lose weight, ranging from dieting to surgery. Increased exercise is a good idea for almost anyone who needs to lose weight. (page 408)

★ **Anorexia nervosa and bulimia.** People suffering from anorexia nervosa deprive themselves of food, sometimes to the point of starvation. They suppress their physiological drives to satisfy other motivations, including self-esteem. People suffering from bulimia alternate between periods of strict dieting and brief but spectacular eating binges. (page 410)

Suggestions for Further Reading

- Capaldi, E. D. (Ed.) (1996). *Why we eat what we eat*. Washington, DC: American Psychological Association. A collection of articles about food selection and the regulation of hunger.
- Logue, A. W. (1991). *The psychology of eating and drinking* (2nd ed.). New York: Freeman. Discusses normal and abnormal eating, including anorexia and bulimia.

Terms

- glucose** the most abundant sugar in the blood (page 403)
- insulin** a hormone that the pancreas releases to increase the entry of glucose and other nutrients into the cells (page 403)
- glucagon** a hormone that the pancreas releases to convert stored energy supplies back into blood glucose (page 404)
- set point** a level of some variable (such as weight) that the body works to maintain (page 405)
- leptin** a hormone released by fat cells; among other effects, it signals the brain to decrease meal size (page 405)
- lateral hypothalamus** an area of the brain that contributes to the control of hunger (page 405)
- ventromedial hypothalamus** an area of the brain in which damage leads to weight gain via an increase in the frequency of meals (page 405)
- paraventricular hypothalamus** an area of the brain in which damage leads to weight gain via an increase in the size of meals (page 405)
- obesity** the excessive accumulation of body fat (page 406)
- anorexia nervosa** a psychological condition in which a person refuses to eat adequate food and steadily loses weight (page 410)
- bulimia** a condition in which a person alternates between self-starvation and excessive eating (page 411)

Answers to Concept Checks

- Insulin levels are higher in the middle of the day (LeMagnen, 1981). As a result, much of the food you eat is stored as fats and you become hungry again soon. Late at night, when insulin levels are lower, some of your fat supplies are converted to glucose, which enters the blood. (page 405)
- By keeping their fat levels very low, they also keep their leptin levels low, and leptin is one of the triggers for puberty. (page 406)
- The set point has increased. (page 406)

Web Resources

Hunger and Eating

www.csun.edu/~vcpsy00h/students/hunger.htm

This student essay is about the biological, learning, and cognitive factors that affect hunger, obesity, and eating disorders.

Eating Disorders

www.psych.org/public_info/eating.html

In the *Let's Talk Facts* series, the American Psychiatric Association describes eating disorders, their major symptoms, what we know about causes, and treatment.

Sexual Motivation

What causes sexual arousal?

What sexual customs are prevalent in our society?

What accounts for some of the variations in sexual practices?

For most of us, sexual activity does not occupy a major fraction of the average day, but thinking about sex does. If you have any doubts about people's interest in sex, consult



Sexual customs vary sharply from one society to another. These people play “kiss-a-girl” in traditional Ukrainian costumes while riding galloping horses. The point of the game is ostensibly to develop horseback-riding skills, but clearly that is not the only motivation.

the television listings, popular books and magazines, and films. (In fact, one way that people can tell they are getting old is that they only think about sex most of the time . . . instead of all the time.)

Humans, unlike most other mammalian species, are interested in sex even at times of the month when a woman is unlikely to get pregnant; in fact, we often take measures to prevent pregnancy. Worldwide, most couples stay together not only long enough to rear children but also long after the children are grown up. Sexual motivation can bind people together in powerful and intimate relationships and also sometimes drive them apart.

Sexual motivation, like hunger, depends on both a physiological drive and available incentives. Also like hunger, the sex drive increases during times of deprivation, at least up to a point, and it can be inhibited for social and symbolic reasons, including religious ones.

However, the sex drive differs from hunger in important ways. We do not need to be around food to feel hungry; many people do need the presence of an attractive partner to feel sexual arousal. Eating in public is normal; sexual intercourse in public is not. We need to eat regularly to stay alive; sexual deprivation is not life-threatening.

What Do People Do, and How Often Do They Do It?

Researchers have many reasons for inquiring about the frequency of various sexual behaviors. For example, if we want to predict how fast and how far the AIDS virus is likely to spread in the population, it is important to know how many people are having unsafe sex with a variety of partners.

In addition to the important scientific and medical reasons for investigating sexual behavior, let's admit it: Most of us are curious about what other people do. We would like to know, “Am I normal? Am I doing something wrong or shameful? Am I missing out on something?”

The answer to the first question depends on what we mean by “normal.” If that term means “reasonably common in the population,” then it is hard to think of anything you might be doing—or not doing—that would make you abnormal. People vary enormously in their sexual behavior and interests.

The Kinsey Survey

The first important survey of human sexual behavior was conducted by Alfred C. Kinsey (Figure 11.20), a shy and studious insect biologist who once agreed to teach the biological portion of Indiana University's course on marriage. When he found that the library included very little information about human sexuality, he decided to conduct a survey. What he intended as a small-scale project eventually grew into a survey of 18,000 people.



FIGURE 11.20 Alfred C. Kinsey pioneered survey studies of sexual behavior. As an interviewer he was peerless—he always put people at ease so that they could speak freely, but he was also alert to probable lies. His results should be interpreted cautiously, however, because he did not obtain a random or representative sample of the population.

Although Kinsey had a large sample, it was neither random nor representative. He obtained most of his interviews by going to organizations, ranging from fraternities to nunneries, and trying to get everyone in the organization to talk to him. As a result, he interviewed mostly midwestern European-Americans who belonged to organizations that agreed to cooperate. Furthermore, with any sample, not everyone gives honest and accurate replies about their sexual activities. Consequently, Kinsey's data do not provide a trustworthy estimate of the frequencies of various behaviors in the U.S. population. Nevertheless, he did document that human sexual behavior is extremely variable (Kinsey, Pomeroy, & Martin, 1948; Kinsey, Pomeroy, Martin, & Gebhard, 1953). For example, he found some men and women who had rarely or never experienced orgasm. At the other extreme, he found one man who reported an average of four or five orgasms per day over the preceding 30 years (with a diary) and several women who sometimes had 50 or more orgasms within 20 minutes.

Kinsey found that most people were unaware of the great variation in sexual behavior in the population at large. For example, when he asked people whether they believed that "excessive masturbation" causes physical and mental illness, most said they did. (We now know it does not.) He then asked what would constitute "excessive." For each person, "excessive" meant a little more than what he or she did. One young man who masturbated about once per month said he thought three times per month would be excessive and would cause mental illness. Another man, who masturbated three times per day, said he thought five times per day would be excessive. (In reaction to these findings, Kinsey once defined a *nymphomaniac* as "someone who wants sex more than you do.")

Contemporary Surveys

Kinsey did not even try to interview a random sample of the population, because he assumed that most people would refuse to cooperate and perhaps even take offense at being asked. He may have been right about people in the 1940s; however, his assumptions clearly do not hold true today. In the 1980s and 1990s, researchers identified random samples of the U.S. population and managed to get cooperation from most of the people they approached (Fay, Turner, Klassen, & Gagnon, 1989; Laumann, Gagnon, Michael, & Michaels, 1994).

(Some advice if anyone ever asks you to participate in a sex survey: Do not cooperate until you know who the questioner is. Legitimate researchers are careful to present their credentials to show their affiliation with a research institute. They also take elaborate precautions to guarantee the confidentiality of people's responses. If "researchers" who want to ask you questions about your sex life fail to present their credentials or seem unconcerned about your confidentiality, do not trust them. Be especially wary of sex surveys conducted by telephone. Although a few sex researchers do conduct research by telephone, most use a face-to-face interview. When in doubt, assume that the alleged survey is an obscene phone call in disguise.)

A survey of a random sample of almost 3,500 U.S. adults (Laumann, Gagnon, Michael, & Michaels, 1994) has added greatly to our knowledge of U.S. sexual practices and customs. First, what would people *like* to be doing? Figure 11.21 shows the percentage of men and women who describe various sexual activities as "very appealing." Clearly, the most popular sexual activity is vaginal intercourse. Watching one's partner undress is "very appealing" to many people; so is oral sex. Other possible options lag well behind. Note one ambiguity in people's responses: A response of "very appealing" might mean, "I have frequently enjoyed doing that activity," "I would like to try that activity," or even "I enjoy fantasizing about that activity, although I might decline to do it in real life." (One wonders, for example, about the 13% of men who say they find group sex very appealing.)

Note also that a higher percentage of men than women report an interest in every activity on the list. Other studies have reported similar results. Besides the differences shown in Figure 11.21, men are much more likely than women to masturbate frequently and to look forward to opportunities for casual sex (Oliver & Hyde, 1993). However, in spite of men's desire for more frequent and varied sexual activities, men and women are equally likely to express satisfaction with their current sex lives.

Figure 11.22 shows the number of sex partners during the past year for people of various ages. At all ages, most people report either no partner or just one. The percentage of people having multiple partners is greatest in young adulthood and declines steadily with age.

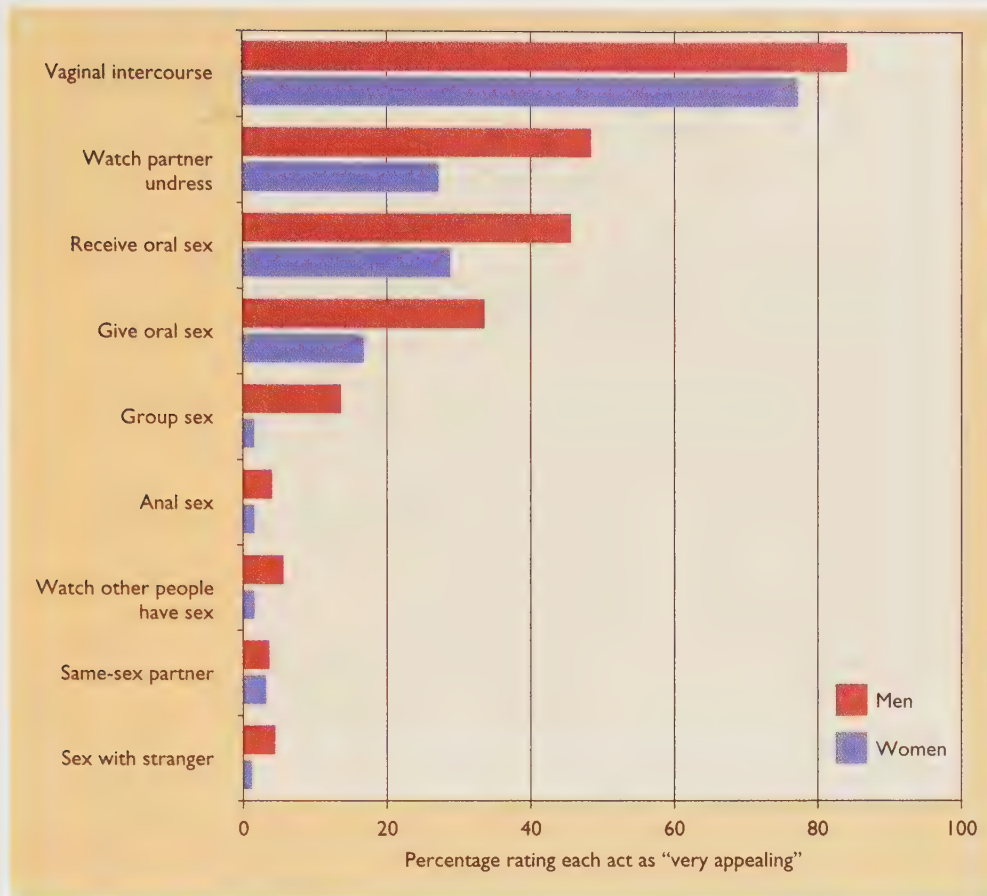


FIGURE 11.21 The percentage of U.S. adults who rate various sexual activities as “very appealing,” as opposed to “somewhat appealing,” “not appealing,” or “not at all appealing.” (Based on data of Laumann, Gagnon, Michael, & Michaels, 1994.)

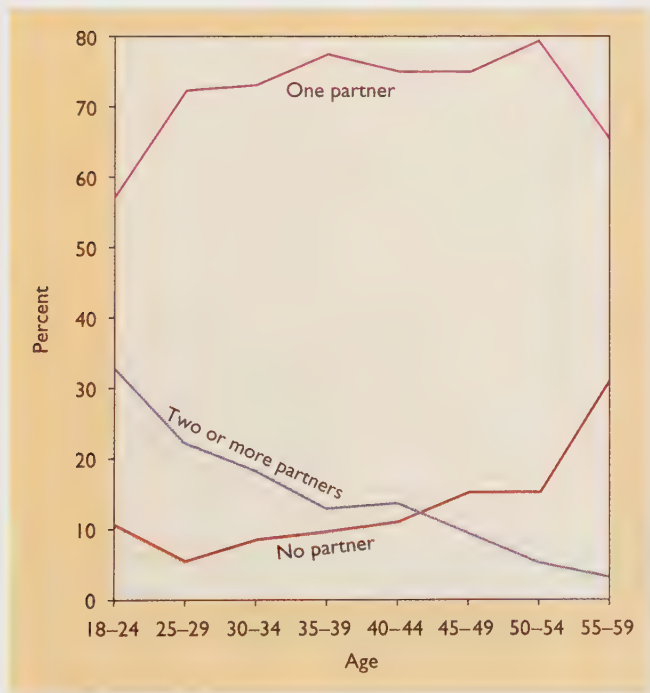


FIGURE 11.22 The number of sex partners in the last 12 months, by U.S. adults of various ages, both sexes combined. (Based on data of Laumann, Gagnon, Michael, & Michaels, 1994.)

The results in Figure 11.22 are cross-sectional, not longitudinal. That is, the 20-year-olds and 50-year-olds grew up in different eras. Figure 11.23 emphasizes these differences among cohorts. People were asked how many sex partners they had had in their lifetime. As Figure 11.23 shows, people in their 40s were more likely to report a large number of partners than were people in their 50s. The reason is that people who were in their 40s at the time of the survey were young and unmarried during the 1970s, an era of a relatively high degree of sexual freedom. People in their 50s were young during a more restrictive era.

In Figure 11.24, you can see differences in current sexual activity reported by people of different ages. The older respondents are less likely than the young ones to report having sex more than twice per week. That trend is probably a real effect of age, not a cohort effect. That is, most 50- to 60-year-olds agree that they are having sexual relations less often than they did when they were younger.

Finally, how many people have a homosexual orientation? You may have heard people announce the figure “ten percent,” as if it were an established fact. That number is derived from Kinsey’s survey from the 1940s and 1950s. Kinsey reported that about 13% of the men and 7% of the women he interviewed reported a predominantly homosexual orientation. The often-quoted figure of 10% is simply the mean of Kinsey’s results for men and women.

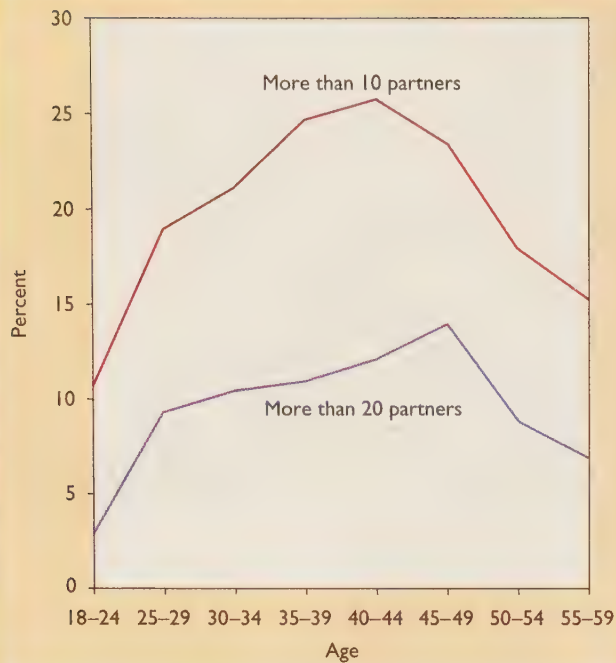


FIGURE 11.23 The percentage of U.S. adults (both sexes combined) reporting more than 10 or 20 sex partners in their lifetime. (Based on data of Laumann, Gagnon, Michael, & Michaels, 1994.)

However, Kinsey's data were based on a very nonrandom sample of the population.

According to a survey of a random sample of 3,500 U.S. adults, 2.8% of men and 1.4% of women describe themselves as having a homosexual (gay or lesbian) orientation (Laumann et al., 1994). As Figure 11.25 illustrates, heterosexuality versus homosexuality is a continuum. Just as many left-handed people perform some tasks with their right hand, and most right-handed people have at least tried various tasks with their left hand, a number of people have at least experimented with both homosexual and heterosexual activities: More than 4% of both men and women report at least one adult homosexual experience. If we expand the scope to "lifetime," including early puberty, 9% of men report at least one homosexual experience.

If you have frequently heard the prevalence of homosexual orientations estimated at 10%, you may be skeptical of the report that only 1–3% of people identify themselves as gay or lesbian. However, three other large surveys of U.S. men reported that 1–2%, 3%, or 6% of U.S. men were either gay or bisexual (Billy, Tanfer, Grady, & Klepinger, 1993; Cameron, Proctor, Coburn, & Forde, 1985; Fay, Turner, Klassen, & Gagnon, 1989). The study reporting the 6% figure had the least satisfactory sampling technique. Surveys in Great Britain and in France have reported a slightly lower prevalence of homosexual orientations, as shown in Figure 11.26 (Spira, et al. 1993; Wellings, Field, Johnson, & Wadsworth, 1994).

The frequencies of various sexual practices vary, of course, among cultures and historical eras. Certainly a sex

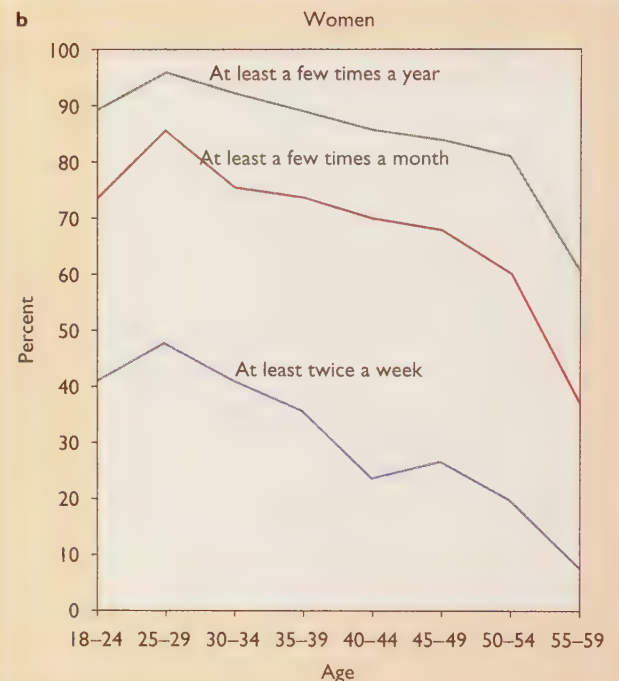
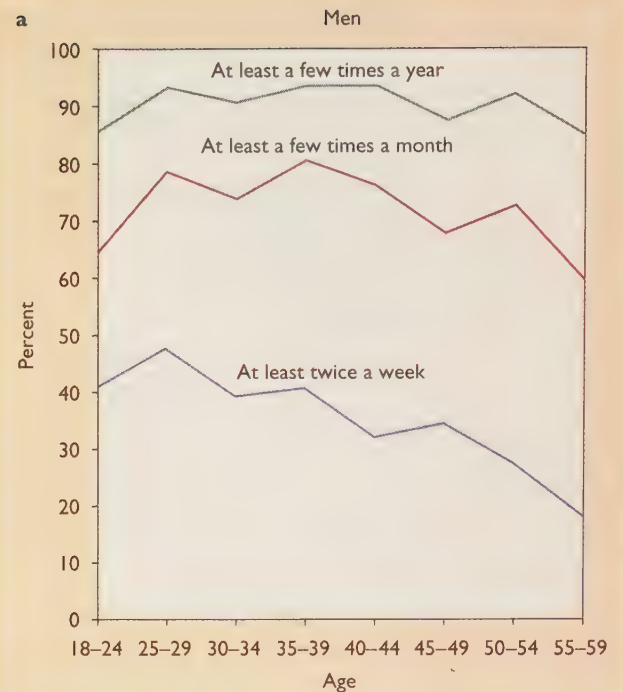


FIGURE 11.24 The percentage of U.S. adults who have had sexual relations with a partner at varying frequencies in the last year. (a) Results for men. (b) Results for women. (Based on data of Laumann, Gagnon, Michael, & Michaels, 1994.)

survey of the United States in 1992 does not apply to other locations or other times. Several studies have reported on the exotic sexual customs of people in certain non-Western

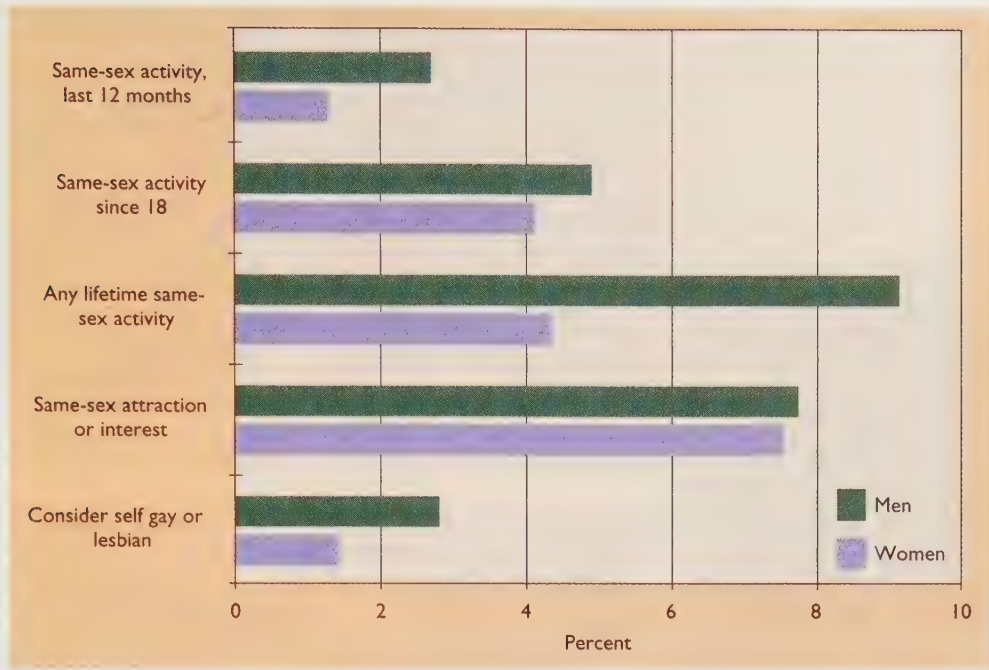


FIGURE 11.25 The percentages of U.S. adults who report sexual activity or interest in sexual activity with others of their own sex. (Based on data of Laumann, Gagnon, Michael, & Michaels, 1994.)

cultures (Davenport, 1977). When we consider how difficult it has been to get reasonably accurate data about U.S. sexual practices, we should be skeptical about studies where anthropologists have interviewed a few people about the sexual customs of their culture. Nevertheless, observable practices (such as dating, marriage, and public display of the human body) vary so strikingly that we can clearly say that much of sexual behavior is the product of learned customs, not just of biological necessity.

Sexual Behavior in the Era of AIDS

During the 1980s, a new factor entered into people's sexual motivations: the fear of **acquired immune deficiency syndrome (AIDS)**, a new and deadly sexually transmitted disease that gradually destroys the body's immune system.

For the AIDS virus to spread from one person to another, it must enter the second person's blood. (The virus does not survive long outside body fluids.) There are three common routes of transmission: transfusions of contaminated blood, sharing needles used for intravenous injections of illegal drugs, and sexual contact. Other contacts between people, even kissing, cannot spread the AIDS virus (unless perhaps both people had a cut on the mouth that was actively bleeding at the time).

An infected male has an estimated 3% chance of transmitting the virus to a female during vaginal intercourse; an infected woman has no more than a 2% chance, probably much less, of transmitting it to a male (Kaplan, 1988). The likelihood of transmission increases if either partner has an open wound on the genitals or if the woman is menstruating. The probability of transmission during anal intercourse

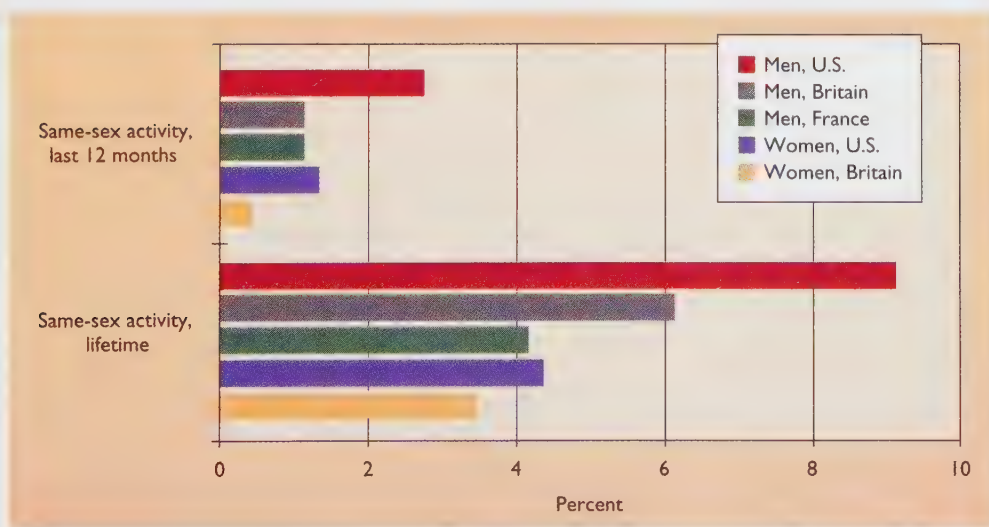
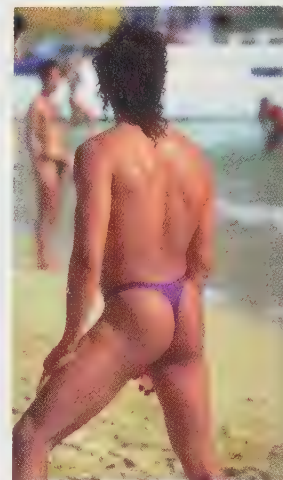


FIGURE 11.26 Comparisons of the results of surveys conducted in three countries, asking people whether they have experienced homosexual activity. (Based on data of Laumann, Gagnon, Michael, & Michaels, 1994; Spira et al., 1993; Wellings, Field, Johnson, & Wadsworth, 1994.)

Human cultures have very different standards regarding public display of the human body, dating, marriage, and premarital sex.



is higher, about 7–10%, because the lining of the rectum is likely to be torn (Kaplan, 1988). These estimates are just approximations, of course.

For generations, people have known how to avoid contracting syphilis, gonorrhea, and other sexually transmitted diseases: Don't have sex with someone who might be infected, or when in doubt, use a condom. If people had consistently followed this advice, we could have eliminated those diseases long ago. The same advice is now offered to combat AIDS, and many people have followed it, though certainly not all. One study of men leaving a homosexual bathhouse found that only 10% were engaging in anal sex without a condom—a far lower percentage than was common before the AIDS crisis (Richwald et al., 1988).

Some heterosexual couples have also become more

cautious, although that increase is not impressive overall. According to one national survey in 1990, only 17% of heterosexuals with multiple sexual partners and 12.6% of heterosexuals with “risky” sexual partners used condoms consistently (Catania et al., 1992). Choosing whether or not to have sex, or how or with whom, is a special kind of decision-making. The continuing spread of the AIDS virus and the continuing number of unwanted pregnancies indicates that many people are still making some unwise sexual decisions (Wyatt, 1994).

Sexual Arousal

Sexual motivation depends on both physiological and cognitive influences. William Masters and Virginia Johnson (1966), who pioneered the study of human sexual response, discovered that physiological arousal during the sex act is about the same in men and women. They observed hundreds of people engaging in masturbation and sexual intercourse in a laboratory and monitored their physiological responses, including heart rate, breathing, muscle tension, blood engorgement of the genitals and breasts, and nipple erection. Masters and Johnson identified four physiological stages in sexual arousal (Figure 11.27). During the first stage, *excitement*, a man's penis becomes erect and a woman's vagina becomes lubricated. Breathing grows rapid and deep. Heart rate and blood pressure increase. Many people experience a flush of the skin, which sometimes re-



AIDS is a preventable disease. By using condoms during sex and by not sharing injection needles with other people, one can greatly decrease the probability of transmitting or receiving the AIDS virus. Advertisements such as this one have prompted many people to change their behavior.

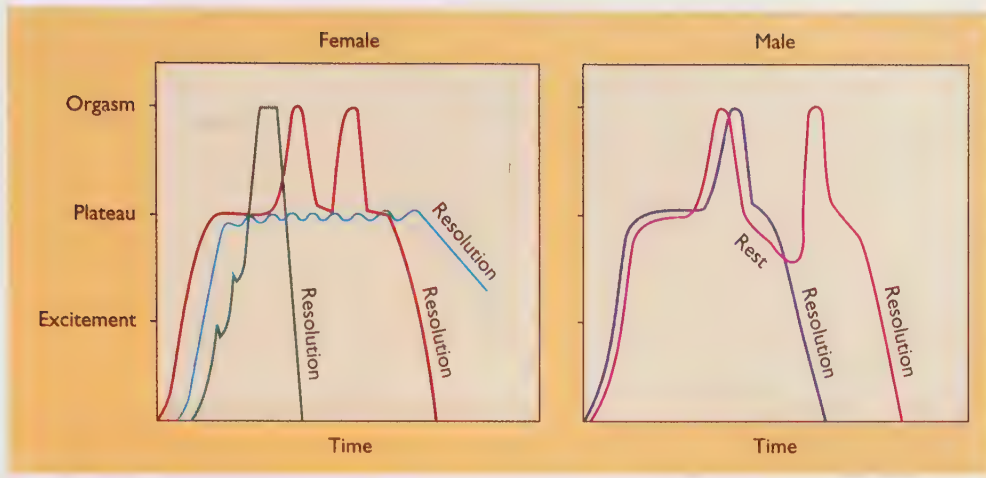


FIGURE 11.27 Sexual arousal usually proceeds through four stages—excitement, plateau, orgasm, and resolution. Each line represents the response of a different individual. (After Masters & Johnson, 1966.)

sembles a measles rash. Women's nipples become erect, and if women have never nursed a baby, their breasts swell slightly. Although this stage is referred to as excitement, it actually requires some degree of relaxation. Nervousness interferes with sexual excitement; so do stimulant drugs (even coffee).

During the second stage, called the *plateau*, excitement remains fairly constant. This stage lasts for varying lengths of time, depending on the person's age and the intensity of the stimulation. During the third stage, excitement becomes intense and is followed by a sudden relief of tension known as *climax* or *orgasm*, which is felt throughout the entire body. During the fourth and final stage, *resolution*, the person returns to an unaroused state.

As Figure 11.27 shows, the pattern of excitation varies from one person to another. During a given episode, a woman may experience either no orgasm at all, a single orgasm, or many. Men do not experience multiple consecutive orgasms, although they can achieve orgasm again following a rest (or refractory) period. In both sexes, the intensity of the orgasm ranges among individuals from something like a sigh to an extremely intense experience.

At any rate, that is the usual pattern. Some people are unable to complete the four stages of arousal. Some men cannot produce or maintain an erection. Others have premature ejaculations, advancing from excitement to orgasm sooner than they or their partners wish. A substantial number of women, perhaps as many as 10%, and a few men stay at the plateau stage without experiencing orgasm. The reasons for such sexual dysfunctions are largely unknown. Some people have physiological disorders; others may have trouble enjoying sexual relations because they feel ashamed.

Sexual Identity and Orientation

Just as hunger includes two major aspects—how much food to eat and which foods to choose—sexual motivation includes two aspects: how frequently to have sex and with

whom. People vary in their sexual preferences, just as they do in their food preferences or any other aspect of life. Why do some people prefer partners of the opposite sex and others prefer partners of their own sex?

Psychologists distinguish between gender identity and sexual orientation. **Gender identity** (or sexual identity) is *the sex that a person regards him- or herself as being*. Ordinarily, people with male genitals have a male identity and people with female genitals have a female identity, although exceptions do occur. **Sexual orientation** is *a person's preference for male or female sex partners (or both or neither)*. People who prefer partners of their own sex have a homosexual (gay or lesbian) orientation. Psychologists do not yet fully understand the causes of sexual orientation.

Influences on Sexual Anatomy

In the earliest stages of development, the human fetus has a “unisex” appearance (Figure 11.28). One structure subsequently develops into either a penis or a clitoris; another structure develops into either a scrotum or labia. The direction this development takes depends on hormonal influences during prenatal development. Beginning in the seventh or eighth week after conception, genetic *male fetuses generally secrete higher levels of the hormone testosterone than do females* (although both sexes produce some), and over the next couple of months the testosterone causes the tiny fetal structures to grow into a penis and a scrotum. In genetic female fetuses, with lower levels of testosterone, the structures develop into a clitoris and labia. Levels of *the hormone estrogen increase more in females than in males* at this time; estrogen is important for female development but has little effect on whether one develops a penis or clitoris, scrotum or labia.

Remember: In humans and other mammals, high testosterone levels produce a male anatomy; low testosterone levels produce a female anatomy. Within normal limits, the amount of circulating estrogen does not determine whether one develops a male or female appearance.

At least one child in two thousand is born with genitals that are hard to classify as male or female. For example, some genetic female fetuses have overactive adrenal glands

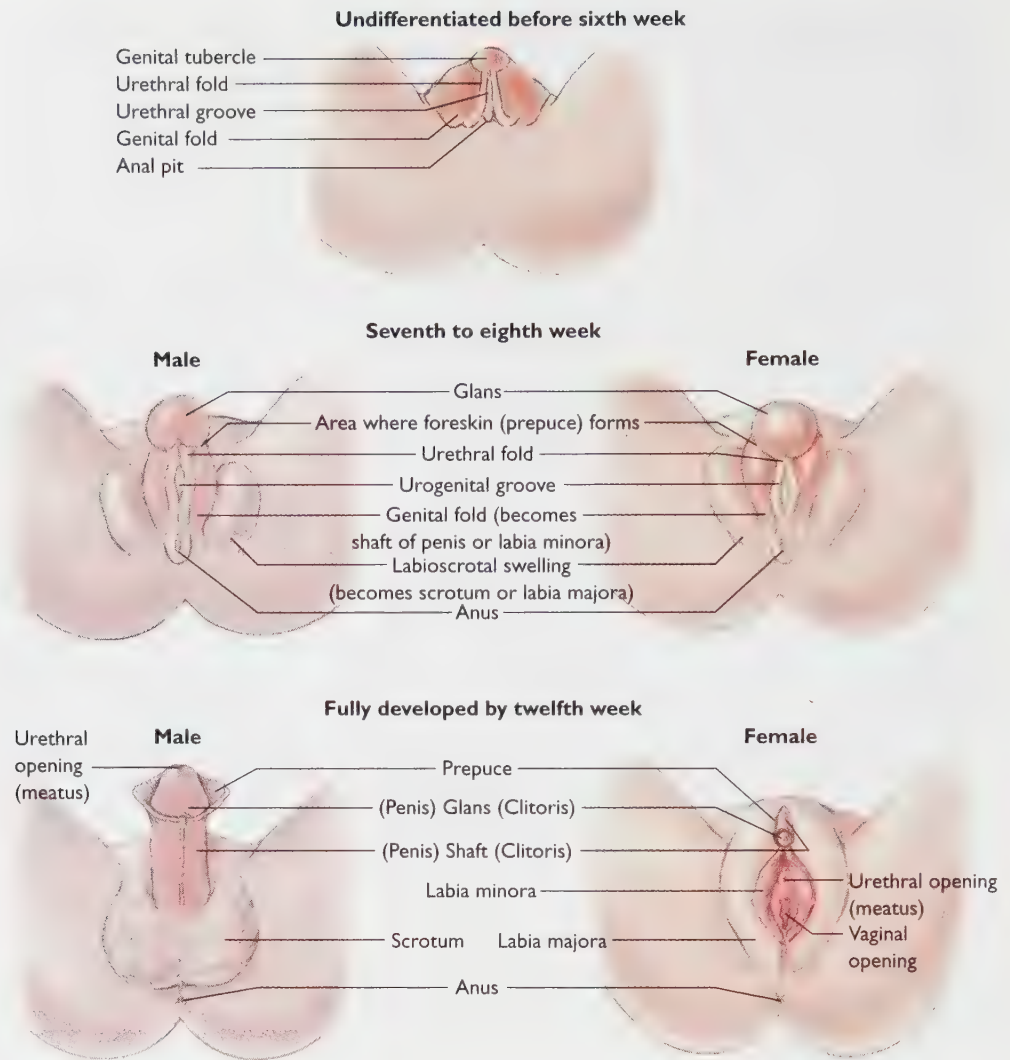


FIGURE 11.28 The human genitals look the same with both male and female fetuses for about the first 6 or 7 weeks after conception (a). Differences begin to emerge over the next couple of months (b) and are well developed at birth (c).

that secrete enough testosterone to partially masculinize development. The genetic female may then develop a sexual anatomy that looks intermediate between male and female (Money & Ehrhardt, 1972). In rarer cases, a genetic male develops an intermediate condition because of a gene that alters hormone receptors (Misrahi et al., 1997). *People with an anatomy that appears intermediate between male and female* are known as **intersexes** or **hermaphrodites** (Figure 11.29).

How should parents and others treat intersexes? The main question is how these people can be most successful and satisfied socially. For decades, the standard medical recommendation has been, when in doubt, call the child female, and perform surgery to make her anatomy look female. This surgery includes creating or lengthening a vagina and reducing the ambiguous penis/clitoris to the size of an average clitoris. In many cases, the additional surgery is unsatisfactory and additional operations are performed later. That is a lot of surgery, but most physicians think it is easier and produces a more normal appearance than any attempts to expand an ambiguous penis/clitoris to the size of a penis.

That recommendation has been based on the assumption that, if a child looks like a girl and is treated like a girl,

she will develop psychologically as a girl. Physicians never had much evidence for that assumption, and some adult intersexes, in spite of their appearance and rearing, develop a male identity. Furthermore, a great many complain that genital surgery—reducing or removing the penis/clitoris—



FIGURE 11.29 External genitalia of a 3-month-old genetic female: This infant was masculinized before birth by excess androgens from the adrenal gland.

left them with greatly reduced sexual excitation and a feeling of mutilation. Finally, they object that, in many cases, physicians lied to them about what surgery they had done and why. The standard, traditional approach in medicine is now being opposed by those who recommend complete honesty with intersexes and avoiding surgery until or unless such people request it themselves as adults (Diamond & Sigmundson, 1997).

CONCEPT CHECK

6. If a human fetus were exposed to very low levels of both testosterone and estrogen throughout prenatal development, how would the sexual anatomy appear? (Check your answer on page 424.)

Influences on Sexual Orientation

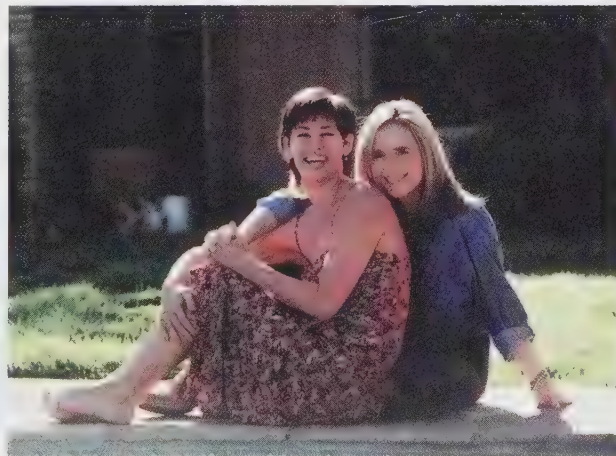
Society's attitudes toward homosexuality have changed repeatedly over time. As far as we know, the ancient Greeks and Romans considered it fairly typical for men to engage in occasional sexual activities with each other as well as with women (Boswell, 1990). (The Greek and Roman writers had little to say about women's sexual interests.) During a later era, Europeans regarded male homosexuality as sinful or criminal. By the early 20th century, the "enlightened" view was that homosexuality was not sinful but "merely" a sign of disease or mental illness. The evidence, however, was based on a badly distorted sample: Psychiatrists and psychologists were acquainted with homosexual men who had sought help for psychological troubles. But homosexual men without psychological problems never consulted therapists, and therefore the therapists did not know about them. Studies of homosexual men who have never visited a therapist revealed that many homosexual men are content and well adjusted (Siegelman, 1974). Consequently, psychologists and psychiatrists today consider a homosexual orientation to be a natural variation in sexual motivation.

The origins and determinants of sexual orientation are not yet well understood. Adult homosexuals often report

that their sexual preference was apparent to them from as early an age as they can remember. They did not choose it voluntarily, and they could not change it, even if they wanted to, any more easily than a left-hander could decide to be right-handed. What causes some people to develop a heterosexual preference and others to develop a homosexual preference?

Most of the available research deals with gay men. Lesbians (women with a homosexual orientation) are less numerous, and many are sufficiently private about their orientation that they become invisible to researchers. The research we do have suggests that genetic factors contribute toward sexual orientation for both men and women. Figure 11.30 shows the results of studies concerning homosexuality in twins and other relatives of adult gays and lesbians (Bailey & Pillard, 1991; Bailey, Pillard, Neale, & Agyei, 1993). Note that homosexuality is more prevalent in the monozygotic (identical) twins than in the dizygotic (fraternal) twins. That trend suggests a genetic influence toward homosexuality, although a gene could hardly be the only factor. (If it were, then 100% of the monozygotic twins would have a homosexual orientation.) Note also that homosexuality is more common among the dizygotic twins than among adopted brothers or sisters. That trend also suggests a genetic factor, although it could also indicate the influence of a factor in the prenatal environment that is shared by twins but not by boys who simply grow up in the same family. If genetic influences contribute to homosexuality, the genes influencing males are different from those influencing females. That is, gay men have a high percentage of gay relatives but not of lesbian relatives; lesbian women have a high percentage of lesbian relatives but not of gay relatives.

If genes affect sexual orientation, they must do so by altering the development of some part of the body, possibly the brain. Other possible influences on sexual orientation, such as prenatal hormones, may also alter brain anatomy. Regardless of the roles of genetics, hormones, and other factors, the evidence suggests a measurable difference in brain anatomy between homosexual and heterosexual men. Let's examine that evidence.



Attitudes toward homosexual relationships have varied among cultures and among historical eras.

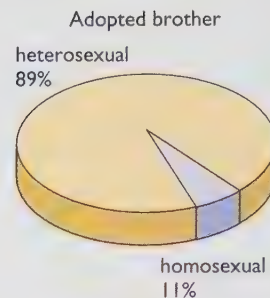
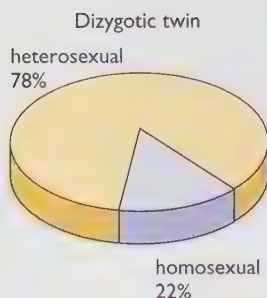
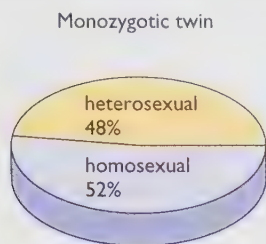
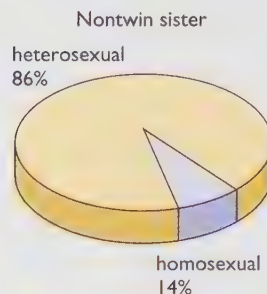
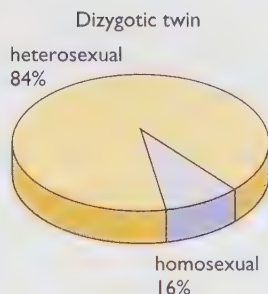
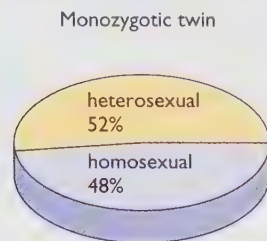
Related to homosexual man:**Related to homosexual woman:**

FIGURE 11.30 The probability of a homosexual orientation is higher among monozygotic twins of adult homosexuals than among their dizygotic twins. The probability is higher among dizygotic twins than among adopted brothers or sisters who grew up in the same family. These data suggest a possible genetic role in the development of sexual orientation. (Based on results of Bailey & Pillard, 1991; Bailey, Pillard, Neale, & Agyei, 1993.)

The Relationship Between Sexual Orientation and Brain Anatomy

Animal studies have demonstrated that one section of the anterior hypothalamus, known to neuroanatomists as INAH3, is generally larger in males than it is in females and that this portion of the brain is necessary for the display of male-typical sexual activity. Its growth is known to depend on prenatal hormones. This, then, is an interesting area to compare in the brains of homosexual and heterosexual men.

Hypothesis INAH3, a particular cluster of neurons in the anterior hypothalamus, will be larger, on average, in the brains of heterosexual men than it is in the brains of homosexual men or heterosexual women.

Method Simon LeVay (1991) examined the brains of 41 young or middle-aged adults (ages 26–59), who had died of AIDS or other causes. AIDS was the cause of death for all 19 of the homosexual men in the study, 6 of the 16 heterosexual men, and 1 of the 6 heterosexual women. No brains of homosexual women were available for study. LeVay measured the sizes of four clusters of neurons in the anterior hypothalamus, including two clusters for

which sex differences are common and two that do not differ between the sexes.

Results LeVay found that three of the four neuron clusters did not consistently vary in size among the groups he studied. However, area INAH3 was, on the average, about twice as large in heterosexual men as it was in homosexual men and about the same size in homosexual men as it was in heterosexual women. Figure 11.31 shows results for two representative individuals. The results probably do not simply reflect the cause of death; among heterosexual men, the size of this brain area was about the same for men who died of AIDS and men who died of other causes.

Interpretation These results suggest that the size of the INAH3 area of the anterior hypothalamus may be related to heterosexual versus homosexual orientation, at least for some individuals. These results are consistent with the idea that genes or prenatal hormones guide brain development, thus altering the probabilities of developing various sexual orientations. However, the results are not entirely conclusive. Conceivably, a homosexual or heterosexual lifestyle might alter brain anatomy instead of the other way around. Also, we do not know whether the people that LeVay studied were representative of other people; certainly, we must await replications on other samples. Finally, the variations in brain structure from one person to another indicate that brain anatomy does not completely control sexual orientation. (The INAH3

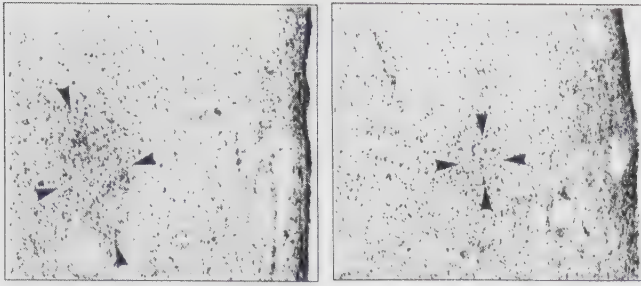


FIGURE 11.31 One section of the anterior hypothalamus (marked by arrows) is larger, on the average, in the brains of heterosexual men (left) than in the brains of homosexual men (right) or heterosexual women (LeVay, 1991). Review Figure 11.13 for the location of the hypothalamus.

nucleus was fairly large in some homosexual men and fairly small in some heterosexual men.)

So, where do all these studies leave us? At this point, the evidence links certain biological factors, including genes and brain anatomy, to male homosexuality, at least for some individuals. We know little, however, about how those biological predispositions combine with experience to produce sexual orientation. Indeed, researchers do not know which kinds of experiences are most crucial. We need to await additional studies before we can draw any confident conclusions.

Uncertainty and tentative conclusions are not unusual in psychology. If you decide to become a psychologist, you will need to get used to the words *maybe* and *probably*. As I pointed out in Chapter 2, psychologists rarely talk about “proving” a conclusion; they merely increase or decrease their confidence in a conclusion.



CONCEPT CHECK

- Most studies find that adult homosexual men have approximately the same levels of testosterone in their blood as heterosexual men of the same age. Do such results conflict with the suggestion that prenatal hormonal conditions can predispose certain men to homosexuality? (Check your answer on page 424.)

THE MESSAGE

The Biology and Sociology of Sex

Sexual motivation at any moment reflects an interplay of biological readiness and the availability of a suitable partner. Similarly, the development of sexual interest and sexual orientation reflects a complex combination of biological predisposition and experiences. Researchers are making progress, but much remains to be learned.

SUMMARY

- Variability in human sexual behavior.** Alfred Kinsey, who conducted the first extensive survey of human sexual behavior, found that sexual activity varies more widely than most people realize. (page 413)
- Prevalence of sexual behaviors.** Modern surveys indicate that most U.S. adults have either one sex partner or none during a given year. Men express a greater interest than women do in varied sexual practices with varied partners. Nevertheless, sexual satisfaction is reported to be about equal for men and women. (page 414)
- Prevalence of homosexuality.** According to several surveys, about 1–3% of U.S. and Western European adults regard themselves as either gay or lesbian. A larger percentage of people, especially men, who regard themselves as heterosexual have experimented with homosexual activities, especially during adolescence. (page 415)
- Sexual arousal.** Sexual arousal proceeds through four stages: excitement, plateau, orgasm, and resolution. For a combination of physiological and motivational reasons, some people fail to pass through all four stages or pass through them more quickly than they wish. (page 418)
- Development of genitals.** In the early stages of development, the human fetus possesses anatomical structures that may develop into either male genitalia (if testosterone levels are high enough) or female genitalia (if testosterone levels are lower). (page 419)
- Homosexuality.** For reasons that are unclear, some people develop a heterosexual orientation and others develop a homosexual orientation. Genetic influences can apparently alter the probability of a homosexual orientation, although these influences alone cannot account for all variations in sexual orientation. On the average, heterosexual and homosexual men differ in the size of one structure in the hypothalamus that contributes to certain aspects of sexual behavior. (page 421)

Suggestion for Further Reading

Laumann, E. O., Gagnon, J. H., Michael, R. T., & Michaels, S. (1994). *The social organization of sexuality in the United States*. Chicago, IL: University of Chicago Press. Provides survey data and also a discussion of the role of sexuality in human life.

Terms

- acquired immune deficiency syndrome (AIDS)** a disease often transmitted sexually that gradually destroys the body's immune system (page 417)
- gender identity** the sex that a person regards him- or herself as being (page 419)
- sexual orientation** a person's preference for either male or female sex partners (page 419)
- testosterone** a hormone present in higher quantities in males than in females (page 419)

estrogen a hormone present in higher quantities in females than in males (page 419)

Answers to Concept Checks

6. A fetus exposed to very low levels of both testosterone and estrogen throughout prenatal development would develop a normal female appearance. High levels of testosterone lead to male anatomy; low levels lead to female anatomy. The level of estrogen does not play a decisive role. (page 421)
7. Not necessarily. The suggestion is that prenatal hormones can alter early brain development. In adulthood, hormone levels are normal, but certain aspects of brain development have already been determined. (page 423)

Web Resources

Sexual Motivation

www.csun.edu/~vcpsy00h/students/sexmotiv.htm

Follow the complex relationships among hormones, cognitions, learning, and culture as they influence sexual motivation.

Achievement Motivation

What motivates some people to work harder than others?

Your 2-year-old nephew is building a stack of blocks. You say, “Here, let me help you,” and you finish stacking the blocks. Will he smile and thank you? Hardly. He is more likely to cry, “I wanted to do it myself!” His goal was not to *have* a tall stack, but to *build* a tall stack.

Say that you are doing something creative yourself—painting a picture, writing a story, playing chess perhaps—something you do moderately well. Someone who is more expert than you says, “Here, let me help you. I see you’re having a little trouble, and I think I can fix it.” How do you react? You might not burst into tears, but you probably resent the help. It was your task, and you wanted to do it yourself.

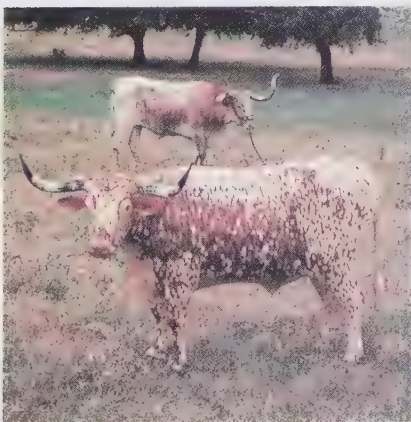
Most of us strive for the joy of accomplishment, some of us more than others. What occupation do you hope to enter after graduation? Have you chosen it because it is the surest way for you to earn a lot of money? Or have you chosen it because it will enable you to take pride in your achievements? Many people have forgone better-paying jobs to take one that will give them a greater feeling of achievement.

Measuring the Need for Achievement

The **need for achievement** is a *striving for accomplishment and excellence*. That sounds like a rather straightforward definition, but it confuses two quite distinct types of motivation (McClelland, Koestner, & Weinberger, 1989). As a rule, when people describe themselves as having a strong achievement motivation, they are referring to an extrinsic motivation. That is, they are drawn by the rewards they have been receiving or expect to receive for various accomplishments. But, as we saw in the first module of this chapter, there is a second kind of need for achievement, a more intrinsic motivation. People with this intrinsic need for achievement may not describe themselves as striving for achievement, but they do take pleasure in accomplishing goals for their own sake. They are likely to persist at a task for a long time and probably develop great skills in the long run. For example, people who spend every spare moment studying and playing chess may be driven by an intrinsic motivation to excel at chess, even if they do not think of themselves as “highly motivated” toward achievement. (They may even think that they are wasting their time.) We shall concentrate here on the intrinsic need for achievement.

The intrinsic need for achievement was first inferred from the performance of schoolchildren. Some children are much more successful in school than others who, as far as we can tell, have equal ability and are equally interested in the rewards that good grades might eventually bring. The same is true in athletics, business, and other aspects of life. Apparently, some people simply try harder than others. If this is true, then we should be able to measure and study this tendency as a personality variable. But how?

If you wanted to determine which workers or schoolchildren were most highly motivated to achieve, what would you measure and how? You could not simply measure how *much* people achieve, because you are trying to explain *why*



The need for achievement includes both extrinsic and intrinsic motivation. The artist who created this wooden cow probably hoped for recognition and money (an extrinsic motivation) but also must have enjoyed the creative process itself (an intrinsic motivation). The inside of the back of the cow folds out to form a desk.

some people achieve more than others. You would need to measure the need for achievement separately from the achievements themselves.

Another way *not* to measure the need for achievement is to ask people whether they are strongly motivated for success. Many people say yes because they believe it is socially desirable to do so. Psychologists measure achievement motivation indirectly, without even telling people what they are measuring.

One of the most popular methods of measuring the need for achievement makes use of the *Thematic Apperception Test* (TAT), which we shall examine again in Chapter 13 (McClelland, Atkinson, Clark, & Lowell, 1953). Investigators show people some pictures and ask them to tell a story about each. Before reading further, try it yourself: Examine the picture in Figure 11.32 and tell a story about it that includes what led up to this scene, what is happening now, and what will happen next.

**TRY IT
YOUR-
SELF**

If you ever take the TAT, after you finish your stories about all the pictures, psychologists will count the number of times you mentioned various themes, including striving for goals and achievements. For example, this story would get a high score on need for achievement:

This girl is taking an important test. First, she went through the test and answered all the items she knew well. Now, she is trying to remember the answer to one of the more difficult questions. She gazes off into the distance, trying to remember everything she has read about this topic. She finally remembers, writes down the correct answer, and gets a perfect score. Later, she goes on to college, becomes a Rhodes scholar, and eventually becomes a famous inventor.

Contrast that story with this one:

This girl is sitting through a very boring class. She is gazing off into the distance, thinking about the party she went to last weekend. As soon as class is over, she goes out and has a good time with her friends.

Such a story would rate a zero on the need for achievement. (It might rate high on other motives, of course.) Your answer on any one picture is not critical; for example, you might tell stories on a few pictures that have nothing to do with achievement and stories rich in achievement themes on the other pictures. What matters is the total over the series.

How well does the Thematic Apperception Test measure need for achievement? The data suggest that it provides a moderately useful measure. Children who score high on the need for achievement generally work harder and get higher grades in school than do children of equal IQs who score low on need for achievement (Khalid, 1991). Need for achievement also correlates reasonably well with various measures of career success (Spangler, 1992). It does not correlate highly with people's self-reports of their "strength of motivation." In other words, the test does seem to tap something related to intrinsic motivation.

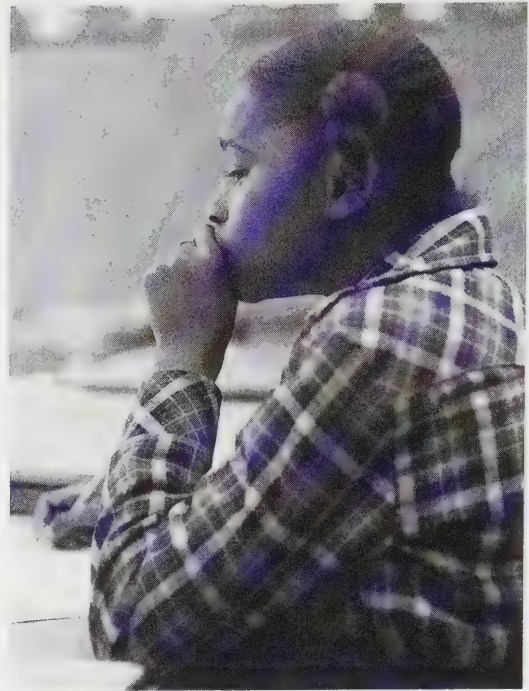


FIGURE 11.32 In the Thematic Apperception Test, each person looks at a series of pictures similar to this one and tells a story about each one. Psychologists count the number of achievement themes mentioned by each person, to measure that person's need for achievement. They can also measure other motivations by counting other kinds of themes.

The Need for Achievement and Setting Goals

Suppose you have a choice of three video games to play. One game is easy; you know you can get a high score on it, but so could anyone else. The second game is more difficult; you are not sure how well you would do. The third is so difficult that you are sure you would lose quickly, as most people do. Which do you choose? Most people prefer the difficult but not impossible game, especially people with a strong need for achievement (Atkinson & Birch, 1978).

People sometimes prefer especially easy or especially difficult tasks if they are dominated by a **fear of failure**, *a concern with avoiding defeat as opposed to gaining victory*. When they try very easy tasks, they avoid failure, although they never achieve any remarkable success. When they try very difficult tasks, they provide themselves with an excuse for failure. Apparently, they would rather fail at an impossible task than run the risk of failing at a realistic task.

People with a strong fear of failure will make a normal effort, or even an extraordinary effort, when given an easy task or when in a relaxed, low-pressure situation. But if they are told, "This is an important test; you are going to be evaluated, so do your best," they expend less effort. In contrast, people with a strong need for achievement make little effort

on an easy task or when the situation puts little pressure on them. When they are told that they are going to be evaluated, they try harder (Nygard, 1982).

When people receive feedback on their performance, such as “you got 82% correct on the first test,” those with a strong need for achievement usually increase their efforts, no matter what the results were. Apparently, they interpret almost any feedback as meaning that they have room for improvement and need to try harder. People with a low need for achievement or a high fear of failure react to feedback by decreasing their efforts. The feedback either tells them that they are achieving their modest goals, or that they are failing and may as well quit (Matsui, Okada, & Kakuyama, 1982).

SOMETHING TO THINK ABOUT

Some people have suggested that our society has become less ambitious and less motivated by achievement than it once was. How could we test that hypothesis? ★

CONCEPT CHECK

8. The new basketball coach at Generic Tech has set up a game schedule for next year. The team will play only opponents that had about average records last year, winning as many as they lost. Does this coach have a high need for achievement or a high fear of failure? (Check your answer on page 431.)

Effective and Ineffective Goals

High goals are especially effective in motivating people with a strong need for achievement, and they also tend to be somewhat effective with most people. (See Figure 11.33.) At the start of the college semester, four young women are asked to state their goals. One is aiming for a straight-A average. Another hopes to get at least a C average. A third plans to “do as well as I can.” A fourth has no set goals. Which student will work hardest and get the best grades?

The student aiming for a straight-A average will do the best, under certain circumstances (Locke & Latham, 1991):

- She must have enough ability for the goal to be realistic. If she has previously struggled just to get passing grades, she will quickly become discouraged.
- She must take her goal seriously. If she casually says she is aiming for straight A's and then never thinks about it again, it will make no difference to her. She can increase her commitment to the goal by stating it publicly. The more people who know about her goal, the harder it will be for her to ignore it.
- She must get some feedback from periodic test scores and grades on assignments to tell her what topics she needs to study harder (Figure 11.34).
- She will be most likely to pursue her goal diligently if she is *not* being paid to achieve it or at least if the payment



FIGURE 11.33 People with a high need for achievement prefer high goals. Research findings indicate that people work hardest when they set high goals for themselves, provided they think they have a realistic chance of succeeding.

does not require an extremely outstanding performance. If she must get all As in order to earn a financial reward, she is likely to become discouraged if it appears that she might get even one B. At that point, she may quit trying. An offer of money helps people to achieve lower goals, but they need intrinsic motivations to pursue high goals.

The same conditions hold for workers (Locke, Shaw, Saari, & Latham, 1981). A very high goal leads to the best performance, provided that the goal seems realistic. A vague “do your best” goal is no better than no goal at all. For a goal to be effective, workers must be committed to achieving it and must receive periodic feedback on their progress. Once they reach their goal, they must be rewarded; otherwise, they will be indifferent toward setting goals later on.

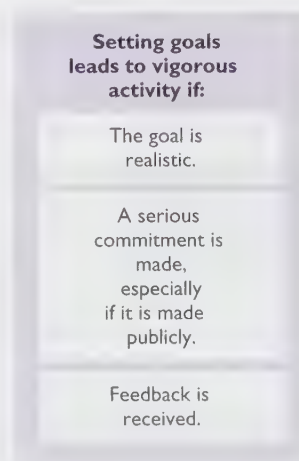


FIGURE 11.34 Conditions for high activity toward achieving goals.

CONCEPT CHECK

9. Under what conditions would people be most likely to keep their New Year's resolutions? (Check your answer on page 431.)

Age and Sex Differences in Need for Achievement

Some people have such a strong need for achievement that they will devote every available moment to an ambitious task they have set for themselves. How does the need for achievement develop, and why does it become stronger in some people than in others?

The Development of Need for Achievement in Childhood

Achievement and success mean different things to different people (Phalet & Claeys, 1993). In some cultures, and for some people within a given culture, the emphasis is on individual accomplishments, such as gaining wealth, fame, and influence. For others, the emphasis is on identification with the group, such as serving one's country or helping one's family. The Japanese culture in particular stresses bringing honor to one's family. In nontechnological societies, success is not defined in terms of jobs and money.

Achievement also has different meanings for people of varying ages. For older people, jobs and earnings become less important goals (Maehr & Kleiber, 1981); hobbies become more important, and even mere competence in taking care of one's own needs can be a source of feelings of accomplishment. Still, even within a sometimes restricted range, people strive for some kind of achievement.

In early childhood, achievement also means something more like competence than prestige. Even children who are only 18 months old clearly show pride in their accomplishments, such as building a tall stack of blocks. By age 2½, they understand the idea of competition; they show pleasure at outdoing someone else, disappointment at losing (Heckhausen, 1984).

Although preschool children show great pleasure when they have completed a task, they seldom appear distressed by their inability to complete it. Heinz Heckhausen (1984) tried to find out how children less than 4 years old would react to failure. He rigged up various contraptions so that a child's stack of blocks would topple or fall through a trap door. He often managed to arouse the children's curiosity, but never their discouragement. He could not find a way to make young children feel that they had failed.

Preschool children are highly optimistic about their own abilities. Even when they have failed a task repeatedly, they confidently announce that they will succeed the next



Preschool children show delight in their successes, but show no clear sign of discouragement after their failures. Perhaps they are ever-confident of their eventual success, or perhaps they simply do not yet understand the concept of failure.

time. An adult asks, "Who is going to win this game the next time we play?" Most preschool children shout "me!" even if they have lost time after time in the past (Stipek, 1984).

Perhaps optimism comes naturally to humans. We quickly learn how it feels to succeed; we learn more slowly what it means to fail. When children enter school, their teachers force them to compare themselves to one another. Within a few years, some children begin to approach tasks with a fear of failure instead of a joyful striving for success (Stipek, 1984).

Psychologists have not yet determined when and why children change their attitudes toward success and failure, but they suspect that teachers unintentionally convey the message, "You probably aren't going to like this or do it very well, but you have to do it anyway." Jere Brophy (1987, p. 190) reports the following quotes from junior-high teachers:

- "If you get done by ten o'clock, you can go outside."
- "This penmanship assignment means that sometimes in life you just can't do what you want to do. The next time you have to do something you don't want to do, just think: 'Well, that's part of life.'"
- "You'll have to work real quietly, otherwise you'll have to do more assignments."
- "This test is to see who the really smart ones are."

Sex Differences in Need for Achievement

Times change. In the United States, before the 1960s, relatively few women sought long-term careers; most expected to quit work when they got married or when they had their first child. However, by the mid-1980s, most women, even mothers of preschoolers, had jobs outside the home (Matthews & Rodin, 1989). Even many older women, who had grown up in an earlier era with much less support for women's careers, reentered the job market (Clausen &

Gilens, 1990). For women as well as for men, work provides an opportunity to gain a sense of accomplishment as well as a source of income.

Nevertheless, industrial-organizational psychologists find that most women still do not advance as rapidly as men do in business. Companies hire women about as frequently as men for their lower-level management positions, but they tend to give men faster promotions and salary increases. When women do reach high levels of leadership, they are generally rated as about equally effective as men at the same level, except in a few settings such as in the military (Eagly, Karau, & Makhijani, 1995).

Why? One possibility is that women place themselves at a disadvantage by interrupting their careers for child care or by refusing promotions that would require moving to another city. However, even women who accept transfers to other cities and who never interrupt their careers still fail to advance as rapidly as men, on the average (Stroh, Brett, & Reilly, 1992).

The seemingly obvious conclusion is that companies continue to discriminate against women, in spite of their claims that they try hard not to do so. Still, there is one more possibility: According to some reports, women, on the average, score lower than men do on need for achievement, as measured by the Thematic Apperception Test. That is, perhaps many women are simply content with a lower rate of advancement, or maybe more men are willing to sacrifice everything else to get ahead.

If men and women really do differ in need for achievement, the explanation must include the fact that society discourages girls and women from setting high goals. Many girls set high goals for themselves while they are in high school but lower their goals within the next few years (Farmer, 1983). In one study, a group of adult women filled out a job-interest questionnaire. The interests they checked most frequently included secretary, elementary school teacher, home economics teacher, and dietitian. Two weeks later, they filled out the same questionnaire again, but this time they were given these instructions:

I want you to pretend with me that men have come of age and that: (1) Men like intelligent women; (2) Men and women are promoted equally in business and the professions; and (3) Raising a family well is very possible for a career woman. (Farmer & Bohn, 1970, p. 229).

After hearing these instructions, the women expressed a significantly increased interest in becoming an author, psychologist, lawyer, insurance salesperson, or physician. They largely lost interest in becoming a secretary, teacher, or dietitian (Farmer & Bohn, 1970). Evidently, women lower their career aspirations because they fear that high ambitions will scare men away, or because they believe businesses will not promote them fairly, or because they fear that a full-time career will interfere with raising a family (Farmer, 1987).

Jobs That Encourage or Discourage Achievement Motivation

Imagine that you are either starting up or reorganizing a company, and you must decide how to divide up the workload. Should you make the jobs challenging and interesting, at the risk that they might be difficult? Or should you make them simple and foolproof, at the risk of being boring?

Your answer will depend on what you assume about the workers and their motivations. According to the **scientific-management approach** to job design, also known as *Theory X*, most employees are lazy, indifferent, and uncreative. Therefore, employers should make the work as foolproof as possible and supervise the workers to make sure they are doing each task the right way, not only to save time but also to avoid injury (Figure 11.35). The employer leaves nothing to chance or to the worker's own initiative (McGregor, 1960).

According to an alternative view, the **human-relations approach** to job design, also known as *Theory Y*, employees like to take responsibility for their work, to enjoy some variety in their job, and to feel a sense of accomplishment (McGregor, 1960). In short, they have a need for achievement. According to this approach, employers should enrich the jobs, giving each employee responsibility for meaningful tasks. For example, a financial services corporation that followed the scientific-management approach would have one employee keep one kind of records, another keep another kind of records, and so on. The same company, reorganized according to the human-relations approach, would put each employee in charge of the services for particular clients, therefore keeping a variety of records, doing different things at various times, and seeing how all the pieces come together. Employees with enriched jobs generally report greater satisfaction (Campion & McClelland, 1991). From the employer's standpoint, the enriched jobs

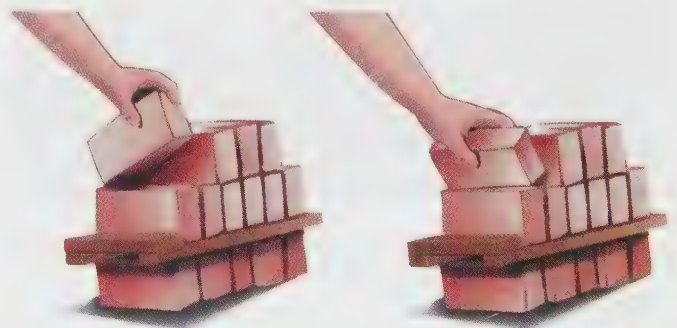


FIGURE 11.35 Psychologists have conducted research to determine the best, safest, most efficient ways to perform even simple tasks. For example, the drawing on the left shows the right way to lift a brick, and the drawing on the right shows the wrong way, according to Gilbreth (1911).

are beneficial in most ways, although they pose two possible disadvantages: It takes longer to train the workers than it would with simpler jobs, and the workers performing enriched jobs expect to be paid more than before!

So, which approach is better? Before you answer, consider an analogy to education: Professor X tells students exactly what to read on which day, what facts to study for each test, and precisely what to do to get a good grade. (This course is analogous to the scientific-management approach; it leaves nothing to the students' initiative.) Professor Y outlines some general issues to be discussed in the course, provides a long list of suggested readings, lets the students control class discussion, and invites students to create their own ideas for projects instead of taking tests. (This course is analogous to the human-relations approach, though perhaps a little more extreme.) Which class would you like better?

My answer would be, "It depends." If I am extremely interested in the topic of the course and I would like to pursue some ideas of my own, then I would prefer Professor Y's course, and I would consider Professor X's course tedious and insulting. But if I am just taking the course to satisfy a requirement and I have no enthusiasm for the topic, I might appreciate the precise structure of Professor X's class.

The same is true of jobs. Some workers, especially the younger and brighter ones, thrive on the challenge of an enriched job. Others, especially those who have been doing a simple job for many years, dislike the insecurity of learning new skills and solving problems on their own (Arnold & House, 1980; Campion & Thayer, 1985; Hackman & Lawler, 1971).

CONCEPT CHECK

10. "I want my employees to enjoy their work and to feel pride in their achievements." Does that statement reflect a belief in the human-relations approach or the scientific-management approach? (Check your answer on page 431.)

THE MESSAGE *What the Need for Achievement Says About Motivation*

Although we sometimes regard hunger as a "typical" motivation, it differs in some striking ways from the need for achievement. For example, people can satisfy their hunger but seldom satisfy their need for achievement. At the end of a big meal, you are quite uninterested in food. In contrast, no matter how marvelous people's achievements are, they seldom lose interest in further achievements. Almost no one reaches his or her life goal and then stops; after reaching a previous life goal, people simply set new goals (Cantor &



People who have already achieved much, such as author Stephen King, seldom rest on their successes. They set new goals and try to achieve even more.

Fleeson, 1994). We would be distressed if we had no further goals. We constantly strive to build a perfect world, but we might not enjoy living in one.

In the words of the Russian novelist Fyodor Dostoevsky (1864/1960, pp. 29, 30): "Man likes to create and build roads; that is beyond dispute. But why does he also have such a passionate love for destruction and chaos? . . . May it not be that he loves chaos and destruction . . . because he is instinctively afraid of attaining his goal and completing the edifice he is constructing? . . . [P]erhaps the only goal on earth to which mankind is striving lies in this incessant process of attaining, or in other words, in life itself, and not particularly in the goal. . . . He feels that as soon as he has found it there will be nothing for him to look for. . . . He likes the process of attaining, but does not quite like to have attained."

SUMMARY

- ✱ **Measurement of need for achievement.** Some people work harder than others because of their strong need for achievement. Need for achievement can be measured by the stories a person tells when looking at pictures in the Thematic Apperception Test. (page 425)
- ✱ **Goal setting.** People with a strong need for achievement prefer to set goals that are high but realistic. Given such a goal, they will work as hard as possible. In contrast, people with a low need for achievement or a strong fear of failure prefer goals that are either easy to achieve or so difficult that they provide a ready excuse for failure. (page 426)
- ✱ **Effectiveness of goal setting.** Almost all of us are motivated to achieve a goal if the goal is realistic, if we make a serious commitment to achieving it, and if we get feedback on our efforts to reach the goal. (page 427)
- ✱ **Achievement motivation in children.** Children begin showing delight in their accomplishments by the age

of 1½. Preschool children are highly optimistic about their own abilities. After they enter school, they learn the meaning of failure and start to show discouragement. (page 428)

★ **Sex differences in need for achievement.** According to some reports, men have a stronger need for achievement than women have, on the average, although some of the results are difficult to replicate. To some extent, women lower their aspirations because they fear that their success may displease men or because they believe that employers will not promote them fairly. (page 428)

★ **Jobs that encourage or discourage need for achievement.** According to the scientific-management approach, jobs should be made simple and foolproof. According to the human-management approach, jobs should be made interesting enough to give workers a sense of achievement. (page 429)

Suggestion for Further Reading

McClelland, D. C. (1987). *Human motivation*. New York, NY: Cambridge University Press. A text by one of the pioneers in the study of achievement motivation.

Terms

need for achievement a striving for accomplishment and excellence (page 425)

fear of failure a preoccupation with avoiding failure, rather than taking risks in order to succeed (page 426)

Answers to Concept Checks

8. The answer depends on how good a team Generic Tech has. If they won most of their games last year and return most of their top players, then the coach has set a very low goal for them, and we can therefore assume that the coach has more fear of failure than need for achievement. If, however, Generic Tech has a mediocre team, then the coach may have set a realistic challenge. We would then assume that the coach has a high need for achievement. (page 427)
9. A New Year's resolution is like any other goal: People are more likely to keep it if it is realistic, if they state the resolution publicly, and if they receive feedback on how well they are doing. (page 428)
10. It reflects the human-relations approach. (page 430)

Web Resources

McGregor—Theory X and Y

sol.brunel.ac.uk/~jarvis/bola/motivation/mcgregor.html

Consider the assumptions and consequences of McGregor's theory of management and motivation: Which management style do you prefer?

David McClelland, Psychologist

fox.nstn.ca/~dmcclel2/psych.html

A brief biography of the famous psychologist, by the not-yet-famous actor.

Motivation

spider.rowan.edu/edlead/marcus/page29.htm

Laurence R. Marcus of Rowan University outlines 10 theories of motivation, including McGregor's, McClelland's, and Maslow's.

Emotions, Health Psychology, and Stress



12

★ MODULE 12.1 **Emotional Behaviors**

Emotion, Decision-Making, and Emotional Intelligence
Excitement and Physiological Arousal

The Role of the Autonomic Nervous System /
The Opponent-Process Principle of Emotions /
The Sympathetic Nervous System and Lie Detection /
THE EFFECTIVENESS OF A POLYGRAPH IN DETECTING LIES:
WHAT'S THE EVIDENCE? / *An Alternative: The Guilty-*
Knowledge Test / Pencil-and-Paper Integrity Tests

Three General Theories of Emotion

The James-Lange Theory of Emotions / The Cannon-
Bard Theory of Emotions / Schachter and Singer's
Theory of Emotions

The Range of Emotions

Producing Facial Expressions / Understanding Facial
Expressions / A Biological Approach to Identifying Basic
Emotions

Happiness

★ MODULE 12.2 **Anger and Violent Behaviors**

Provocations of Anger and Violence

Characteristics of Relatively Violent People

Violence by Mental Patients

Sexual Violence

★ MODULE 12.3 **Health Psychology**

Stress

Selye's Concept of Stress / Posttraumatic Stress
Disorder / Measuring Stress and Its Effects on Health /
Lazarus's Approach to Stress

Stress and Psychosomatic Illness

Heart Disease / Cancer

★ MODULE 12.4 **Coping with Stress**

Coping Styles and Strategies

Monitoring Strategies

The Importance of Predictability and Control /
Inoculation / Social Support / Beliefs as Coping
Strategies

Blunting Strategies

Relaxation / Exercise / Distraction

The *Star Trek* character Mr. Spock is reputed to have emotions but to suppress them almost always. Suppose you are the first astronaut to land on a planet where the inhabitants experience no emotions at all (but who conveniently speak English). They gather around and ask you, their first visitor from Earth, what *emotion* is. What do you tell them?

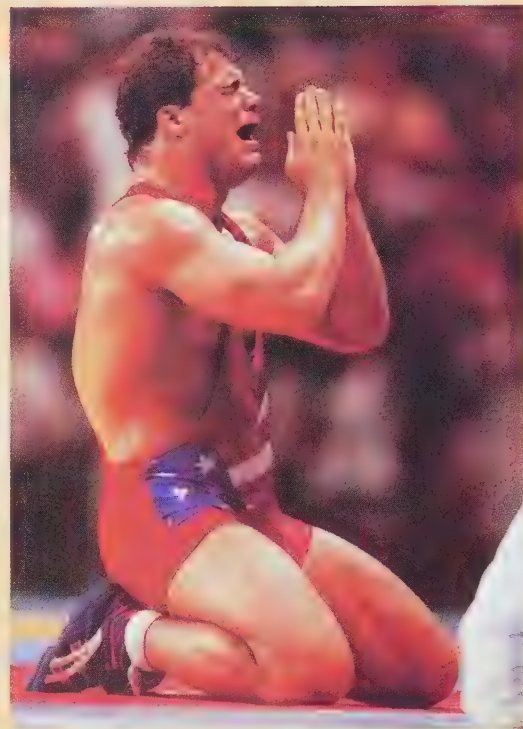
"Well," you might say, "emotion is how you feel when something surprisingly good or surprisingly bad happens to you." "Wait a minute," they reply. "We don't understand these words *feel* and *surprisingly*."

"All right, how about this: Emotions are experiences like anger, fear, happiness, sadness. . . ."

"*Anger, fear* . . . what do those terms mean?" they ask.

Defining such terms would be more difficult than trying to explain *color* to a blind person. Even though blind people cannot experience color, they can determine whether someone else has color vision by showing the appropriate stimuli. Could you set up a similar test that would allow emotionless people to determine whether someone else was experiencing an emotion? If so, what sort of test could you use? The problem is that *color vision* has a well-established meaning, whereas *emotion* has an imprecise meaning that is defined mostly by example.

In this chapter, we shall consider what psychologists have learned so far about emotions, while leaving some major questions unanswered. We begin with general theories and principles. Later, we turn to the role of emotions in health and how people cope with the emotions associated with stress.



When we experience emotions, we are in some way moved. (*Movement is the root of emotion.*) Our feeling of disturbance is a positive or negative excitement; sometimes we cry for joy, sometimes in grief.

Emotional Behaviors

What causes us to feel one emotion or another?

How many kinds of emotion do people have?

The philosopher Paul Griffiths (1997, p. 14) wrote, “My central conclusion is that the general concept of emotion is unlikely to be a useful concept in psychological theory.” He compares the concept of emotion to ancient astronomy’s concept of “superlunary objects,” which referred to all objects beyond the orbit of the moon: The items that fit the concept are too diverse to form a useful category.

If you would like to defend the category *emotion*, first try to define it. When you find it difficult to define (as you will), you may object, “Even if I can’t define it in so many words, I can recognize it when I see it.” Can you? When you swat at a housefly and it flies away, is it frightened? If you poke at a beehive and the bees attack you, are they angry? If fear and anger refer only to the behaviors, then, yes, the fly is frightened and the bees are angry. But if emotions are experiences, like what you and I feel when we experience fear and anger, then it is difficult—maybe even impossible—for us to know whether insects have emotions.

Never mind insects; what about humans: Is boredom an emotion? interest? pain? courage? confidence? sexual desire? How do we decide what is an emotion and what is not?

Emotion is often a slippery concept, difficult to define and difficult to measure. It is, nevertheless, so important that psychologists can hardly ignore it. The cautious approach is to anchor our discussion as carefully as possible in observable, measurable behaviors.

Emotion, Decision-Making, and Emotional Intelligence

When drawing conclusions, you are generally advised to look at the evidence “calmly, rationally, unemotionally.” The *Star Trek* character Mr. Spock, the personification of this advice, is extremely logical and rarely emotional. If emo-

tions only get in the way of intelligent decision-making, why do we have them at all?

The answer is that, except for extremely intense emotions, they don’t get in the way. In fact, the capacity to feel emotion may even be necessary for good decision-making. Antonio Damasio (1994) described some patients who suffered severely impoverished emotions following damage to parts of the prefrontal cortex. One was the famous patient Phineas Gage, who survived a freak accident in 1848 when an iron bar shot through his head, damaging his prefrontal cortex. Nearly one-and-a-half centuries later, researchers examined his skull (which is still on display in a Boston museum) and reconstructed the route that the bar must have taken through Gage’s brain (Damasio, Grabowski, Frank, Galaburda, & Damasio, 1994). As you can see in Figure 12.1, the accident damaged a portion of his prefrontal cortex. The result, according to medical reports of the time, was that Gage showed little emotion and also seemed to lose his former values. He had formerly been a conscientious worker, but he became unreliable in both his work and his personal habits. He ignored the usual polite restraints when making sexual advances and in his use of profane language. He followed each whim of the moment, unable to carry out any long-term plans.

A parallel case in more recent times is a person known to us as “Elliot” (Damasio, 1994). Elliot, too, suffered damage to his prefrontal cortex, as a result of surgery to remove

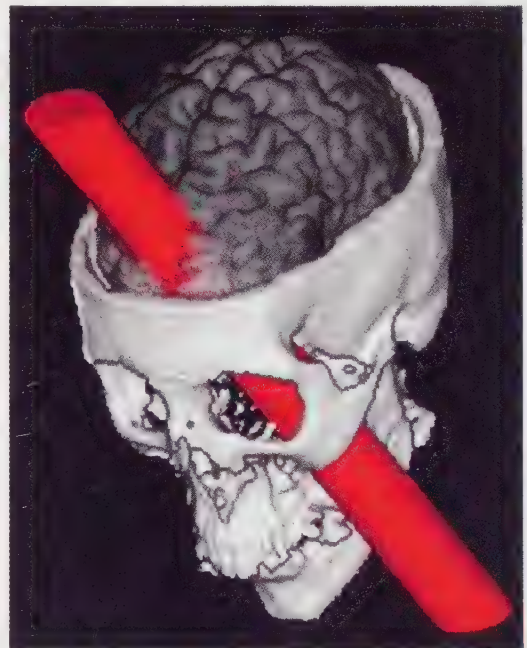


FIGURE 12.1 In the 1990s, researchers used modern technology to reconstruct the path that an iron bar must have taken through the brain of Phineas Gage, who survived this injury in 1848. The damage impaired Gage’s judgment and decision-making. (From Damasio, Grabowski, Frank, Galaburda, & Damasio, 1994.)